

Dynamic Screen Space Textures for Coherent Stylization

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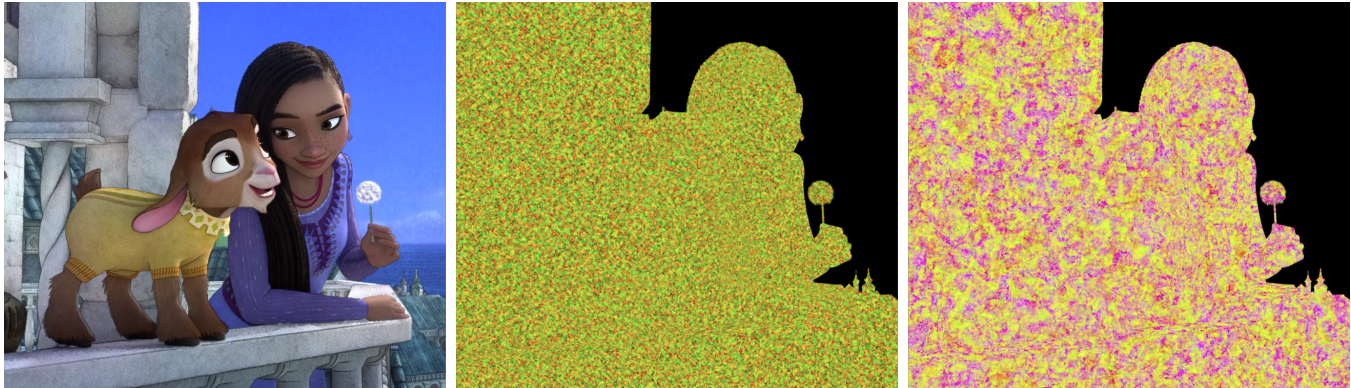


Figure 1: Two of the dynamic screen space textures used to create the stylized look of “Wish”. ©2024 Disney

ABSTRACT

Achieving a watercolor look was an important goal for the style of Walt Disney Animation Studios’ “Wish”, and screen space textures were critical for achieving this, for example to convey a sense of the watercolor paper texture. However, using traditional screen space textures would have resulted in a distracting shower-door effect where the animation appears to swim through the texture. Our novel dynamic screen space textures overcame this problem by tracking animation and camera movement while maintaining the screen space qualities of the texture.

CCS CONCEPTS

• Computing methodologies → Rendering; Ray tracing.

KEYWORDS

ray tracing, animation, global illumination

ACM Reference Format:

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1 INTRODUCTION

The watercolor look in “Wish” is the result of compositing various texture layers including a watercolor paper texture and several additional noise textures that add the impression of brush strokes, edge breakup, pigment pooling, and other stylization effects.

Our textures faced two seemingly conflicting requirements:

- The textures must maintain their screen space feature size and camera-facing orientation.
- The textures must track camera movement and animation, including when objects deform, rotate, or move towards or away from the camera.

Under animation, the textures should feel like they are part of the characters and environment, but when the animation stops, the textures should feel like they are part of the watercolor medium, unifying the elements in the scene.

The closest previous technique is MetaTextures [Curtis et al. 2019], which blends textures at different UV scales to maintain screen size, and applies skew compensation to align the textures to the screen, however some perspective is still visible. Also, MetaTextures require high-quality, seamless UVs.

Our *dynamic screen space textures* are able to maintain camera-facing orientation and screen space scale across static or deforming objects, including objects that do not have UVs, as well as aggregate geometry like hair, fur, or foliage.

2 DYNAMIC SCREEN SPACE TEXTURES

We start with an artist-specified exemplar texture and a desired screen space scale. We then determine 3D anchor points in the scene and map a camera-facing texture instance to each point.

For rigid or static geometry, we use a virtual 3D lattice to define the anchor points, in a similar manner to Perlin noise. An example lattice is visualized in Figure 2.

If the lattice were used with deforming objects, the animation would appear to swim through the textures. To avoid this, instead of using lattice points we choose anchor points from among each object's control vertices (CVs) which intrinsically track the animation. The points are carefully selected to provide approximately uniform spacing at each of a set of progressively higher densities. We refer to this approach as *progressive points*, which we describe in more detail in Section 3. Results comparing the lattice and progressive-points approaches are shown in Figure 3.

At each shading point, we look up the 8 nearest 3D anchor points, perform a camera projection of each texture to the shading point, and blend the results. During the camera projection, we add a 2D rotation to align the camera's up direction with the object's local (3D) up direction. An additional 2D rotation can be specified by the artist to control the orientation of texture.

To maintain a consistent point spacing and texture scale across camera moves, we define the point sets at multiple power-of-two scales, and perform the lookup on each of the two nearest scales.

The anchor point spacing is independent of the texture scale, but affects the perceived quality. If the anchor points are too close together, the texture details will be broken up; and if the anchor points are too far apart, the texture may appear to slide as objects deform or rotate. We provide an artist control for the target point spacing, but this is not something that typically needs to be adjusted once a good default is determined.

To minimize ghosting and preserve contrast in the blended regions, we use histogram-preserving blending with exponentiated weights [Burley 2019]. In theory, we must evaluate and blend 16

texture samples for each shading point (8 for each of the two nearest scales). However, with exponentiated weights, many of the 16 textures will have weights close enough to zero that they can be ignored, meaning that only a few texture lookups are typically needed at each shading point.

3 PROGRESSIVE POINTS

A deformed object's CVs intrinsically track the deformations, and typically have more than enough density to achieve the desired point spacing. Our task is to choose an evenly-spaced subset of the CVs for each of the required power-of-two scales, and to store these points in a way that the nearest points can be efficiently determined at shading time.

We construct our progressive point sets for each deforming surface at render initialization time. To choose the points, we form a set of power-of-two sized grids over each deforming surface. For each grid cell, we choose the CV that is closest to the center of the cell. This is done with a single pass over the CVs, updating each containing grid cell if the CV is closer to the center than the previously-closest one.

In order for the point set to be consistent frame-to-frame, we build the grids using each CV's reference position (a.k.a. PRef or "rest pose"), and construct a kd-tree per grid for efficient nearest-point search during shading.

For non-deforming surfaces, lattice points are still preferred as they are more efficient, and they work for surfaces where using the CVs would not provide sufficient point density.

4 CONCLUSION

Dynamic screen space textures played a critical role in achieving the stylized watercolor look of Walt Disney Animated Studios' "Wish". The textures were applied to virtually every element in every shot of the film, and the production proceeded with few issues, and minimal burden on the artists. This technology is expected to be used in all of our stylized rendering projects going forward.

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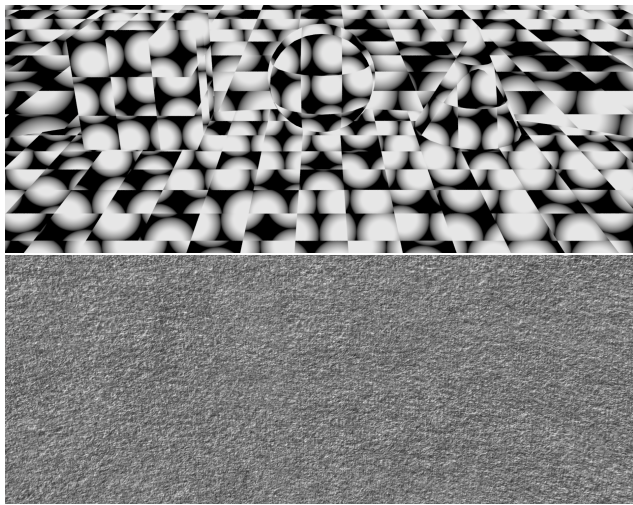


Figure 2: Dynamic screen space texture on simple 3D scene. Top: a circle gradient texture is applied with no blending to reveal the lattice and the 3D geometry; note the change in lattice scale near the middle of the image. Bottom: a watercolor paper texture is applied to the same scene; enabling blending completely hides the lattice and geometry, and produces a consistent screen space texture result across the image. ©2024 Disney

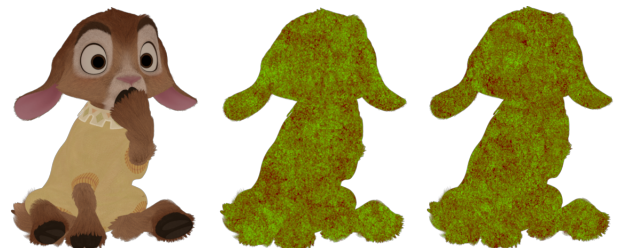


Figure 3: Lattice-based points (middle) produce a good result on a deformed character in a still frame, but would swim during animation. Progressive points (right) produce a good result under still or animated deformation. ©2024 Disney